

Fabrizio Romanelli

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About

Senior Technical Leader and Research Engineer with 20 years of sustained impact spanning industrial robotics, physics-based simulation, and deep learning. Expert in bridging the sim-to-real gap for complex electromechanical systems by leveraging advanced visual perception, synthetic sensor data generation, and World Models. Proven track record in designing real-time robot control frameworks, multibody kinematics, and contact-rich collision-avoidance algorithms within production engineering environments (Comau, IIT, Almagora). Technical vision leader and experienced software manager adept at coordinating cross-functional teams of 10+ people, deploying scalable AI pipelines, and architecting advanced simulation-driven workflows for next-generation intelligent systems.

Education

PhD Computer Science, Control and Geoinformation – University of Rome Tor Vergata	2020 – 2024
MSc Computer Science (with distinction) – University of Rome – Tor Vergata [with distinction, 110/110 cum laude]	2003 – 2005
BSc Computer Science – University of Rome – Tor Vergata	1998 - 2003

Professional experiences

AI & Robotics Engineer – Almagora

Architected and designed project proposals integrating World Models and generative AI to systematically predict physical behaviors and close the sim-to-real gap in autonomous systems.

Designed and developed industry-grade AI solutions for advanced visual object detection (YOLOv8) and robust perception in highly cluttered, non-deterministic operational environments.

05.2024 – present

Engineered and deployed scalable, enterprise-level solutions leveraging Large Language Models (LLMs) to extract automated project insights, requirements, and technical risk analysis.

Served as a technical mentor for engineering teams, driving best practices in object-oriented programming (OOP), software versioning, and agile deployment pipelines.

Senior Software Engineer – Agrivolt

Developed production-grade software stacks for agricultural robotics, utilizing deep learning techniques for high-fidelity synthetic sensor data generation and real-time image segmentation.

Engineered resilient navigation algorithms capable of handling highly variable physical terrains and dynamic environmental contacts.

06.2021 – 05.2024

Collaborated directly on corporate business strategy, mapping technical simulation and AI capabilities to long-term commercial product roadmaps.

Researcher – CNR-ISTP

Awarded a research fellowship focused on the Electron Cyclotron Resonance Heating (ECRH) system for the DTT (Divertor Tokamak Test facility) international nuclear fusion project.

12.2022 – 05.2024

Designed the complex control architecture, real-time remote monitoring systems, and automated power supply management systems for high-power gyrotrons.

Training Specialist on industrial robotics – IN.SI.

Conducted advanced technical training on industrial robot control systems, rigid-body kinematics, multi-body dynamics, and fieldbus-based PLC interfacing.

03.2021 – 12.2023

Developed high-fidelity interactive simulations to model, test, and program complex kinematics workflows for Universal Robots and ABB platforms.

Senior Cloud Engineer – Applaudart	03.2019 – 01.2021
<i>Led the core backend development team architecting high-performance cloud computing systems on Amazon Web Services (AWS), utilizing Lambda, DynamoDB, ElasticSearch, and AppSync.</i> <i>Specialized in the integration of highly complex, enterprise-class distributed systems with automated CI/CD pipelines to ensure scalable out-of-simulation workflows.</i>	
Robotics Specialist – University of Rome Tor Vergata	01.2020 – 05.2020
<i>Designed a distributed, low-latency ROS 2 software architecture optimized for resilient perception in autonomous mobile systems.</i> <i>Investigated state-of-the-art VSLAM and deep learning methodologies for physics-based synthetic sensor data generation.</i> <i>Developed high-level C++ wrappers for RGB-D camera drivers, integrating and benchmarking ORBSLAM2 optimization models across monocular, stereo, and depth-sensing arrays in real-world environments.</i>	
Senior Software Engineer – Istituto Italiano di Tecnologia	12.2017 – 02.2019
<i>Core member of the Advanced Robotics Research Line; supervised engineering teams and postgraduate students in developing real-time control frameworks for legged robots.</i> <i>Designed modular software architectures for robotic locomotion and multi-DoF manipulation, ensuring high-predictive accuracy during contact-rich physical interactions.</i> <i>Managed and maintained advanced CI/CD regression testing pipelines on GitLab; integrated open-source packages directly into physical robot-in-the-loop (HIL/SIL) testing environments.</i>	
Software Development Manager – Comau	03.2016 – 10.2017
<i>Directed the technical software design, architecture, and deployment of Comau's next-generation industrial robotics platform across PC, mobile, and real-time controller environments.</i> <i>Translated ambiguous real-world electromechanical behaviors into tractable, safe, and robust motion control models.</i> <i>Led cross-functional engineering teams, fostering technical development, cross-department alignment, and execution of complex project milestones.</i>	
Software Engineer Specialist in Motion Planning and Robotics – Comau	03.2007 – 02.2016
<i>Designed, coded, and validated advanced motion control algorithms, kinematic solvers, and trajectory planners for Comau industrial manipulators using C/C++ on Real-Time Operating Systems (RTOS).</i> <i>Active technical contributor to high-profile EU Projects (ROBOFOOT, SMERobotics, X-Act); developed the C4G/C5G Open and Open Realistic Robot Library to support motion planning on open Linux platforms.</i> <i>Led the internal eMotion project, directly optimizing multi-body dynamic models to maximize robot velocity profile smoothness while reducing mechanical stress.</i> <i>Engineered engineering-grade software solutions for advanced Conveyor Tracking and contact/collision avoidance in overlapping Interference Regions, managing complex flexible assemblies and multi-robot workspace safety.</i>	
Software Engineer Specialist in Motion Planning and Robotics – ICAPGroup	10.2006 – 03.2007
<i>Integrated highly automated, safety-critical robotic cells for the automotive manufacturing sector using Comau and ABB platforms.</i> <i>Designed specialized computer vision systems using deterministic programming languages and programmed complex Allen-Bradley PLC and Pilz hardware-in-the-loop security configurations.</i>	

Activities, achievements & certifications

- Associate Editor Online Journal of Robotics & Automation Technology (OJRAT)
- Editorial Board Member Sciencefather
- Graduated with highest honors from Rome University of Tor Vergata
- Reviewer: IEEE Sensors Journal, ICRA 2023, IEEE Journal of Radio Frequency Identification, Elsevier Computers & electrical engineering, Elsevier Digital signal processing
- Second prize winner at EUnited Robotics Technology Transfer Award 2014

- Finalist at EUROP/EURON Technology Transfer Award 2009
- Second prize at Altran 'Campus' Award 2006
- Fundamentals of accelerated computing with CUDA Python – NVIDIA certification
- Disaster risk monitoring using satellite imagery – NVIDIA certification
- Getting started with AI on Jetson Nano – NVIDIA certification
- Building Video AI applications at the edge on Jetson Nano – NVIDIA certification
- Develop, Customize, and publish in Omniverse with extensions – NVIDIA certification

Skills and competences

Core Research & Systems: Physics-Informed ML, World Models, Sim-to-Real Alignment, Multi-Body Dynamics, Synthetic Data Generation, Contact & Friction Modeling, Sensor Fusion.

Programming Languages: Python, C, C++, MATLAB, JavaScript, Perl, FORTRAN.

Frameworks, Libraries & Tools: PyTorch, TensorFlow, ROS/ROS2, OpenUSD, OpenCV, NumPy, Pandas, GitLab CI/CD, AWS Cloud Infrastructure.

Languages: Italian (Native), English (Full professional proficiency), Spanish (Full professional proficiency), French (Limited working proficiency).

Selected publications

Romanelli, F., & Martinelli, F. (2023). *Synthetic sensor measurement generation with noise learning and multi-modal information*. IEEE Access, doi: 10.1109/ACCESS.2023.3323038.

Romanelli, F., & Martinelli, F. (2023). *Synthetic Sensor Data Generation Exploiting Deep Learning Techniques and Multi-Modal Information*. IEEE Sensors Letters, vol. 7, no. 7, pp. 1-4, July 2023, Art no. 7003404, doi: 10.1109/LSSENS.2023.3290209.

Romanelli, F., Martinelli, F., & Mattogno, S. (2023). *Resilient simultaneous localization and mapping fusing Ultra Wide Band range measurements and Visual Odometry*. Journal of Intelligent & Robotic Systems. (Accepted for publication)

Navone, A., Romanelli, F., Ambrosio, M., Martini, M., Angarano, S., & Chiaberge, M. (2023). *Lavender Autonomous Navigation with Semantic Segmentation at the Edge*. arXiv preprint arXiv:2309.06863.

Bianchi, L., Carnevale, D., Del Frate, F., Masocco, R., Mattogno, S., Romanelli, F., & Tenaglia, A. (2023). *A novel distributed architecture for unmanned aircraft systems based on Robot Operating System 2*. IET Cyber-Systems and Robotics, 5(1), e12083.

Martinelli, F., Mattogno, S., & Romanelli, F. (2023). *A resilient solution to Range-Only SLAM based on a decoupled landmark range and bearing reconstruction*. Robotics and Autonomous Systems, 160, 104324.

Bianchi, L., Carnevale, D., Masocco, R., Mattogno, S., Oliva, F., Romanelli, F., & Tenaglia, A. (2023, May). *Efficient visual sensor fusion for autonomous agents*. In 2023 International Conference on Control, Automation and Diagnosis (ICCAD) (pp. 01-06). IEEE.

Romanelli, F., Martinelli, F., & Di Giampaolo, E. (2022). *Robust simultaneous localization and mapping using range and bearing estimation of radio ultra high frequency identification tags*. IEEE Transactions on Control Systems Technology, 31(2), 772-785.

Fenucci, A., Indri, M., & Romanelli, F. (2014, September). *A real time distributed approach to collision avoidance for industrial manipulators*. In Proceedings of the 2014 IEEE Emerging Technology and Factory Automation (ETFA) (pp. 1-8). IEEE.

Michieletto, S., Tosello, E., Romanelli, F., Ferrara, V., & Menegatti, E. (2014). *ROS-I interface for COMAU robots*. In Simulation, Modeling, and Programming for Autonomous Robots: 4th International Conference, SIMPAR 2014, Bergamo, Italy, October 20-23, 2014. Proceedings 4 (pp. 243-254). Springer International Publishing.

Bascetta, L., Ferretti, G., Magnani, G., Rocco, P., Abba, F., Gerio, G. P., & Romanelli, F. *Towards an industrial implementation of the walk-through programming technique for robotic manipulators*. ICRA 2012 IEEE International Conference on Robotics and Automation.

Patents

- Method for controlling at least two robots having respective working spaces including at least one region in common - US US20130110288 A1 · Issued May 2, 2013
- Robot system - EU EP 2194434 (A1) · Issued Dec 5, 2008 and US US 8,412,379 B2 · Issued Apr 2, 2013